



# MUST 'A' MATHS

SM015

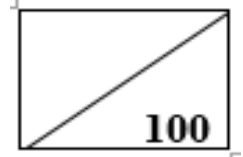
SESI 2025/2026

KOLEJ MATRIKULASI NEGERI SEMBILAN



Name: \_\_\_\_\_

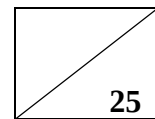
Class: \_\_\_\_\_



2 Hours

## SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.



1. Find the following limits, if they exist.

(a)  $\lim_{x \rightarrow 3} \frac{\sqrt{x-1} - \sqrt{2}}{x-3}$ . [3 marks]

(b)  $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+5}}{2-x}$ . [3 marks]

(c)  $\lim_{x \rightarrow 2^-} \frac{|x^2+x-6|}{2x-x^2}$ . [3 marks]

2. Find  $\frac{dy}{dx}$  for the following equations.

(a)  $y = xe^2 - \ln(2x+1)^{x^2}$  [5 marks]

(b)  $y = x^3 \sin^2 3x$  [4 marks]

3. Given  $f(x) = x^4 - 2x^3 + 10$ .

(a) Determine the interval where  $f(x)$  is concave upward or concave downward.

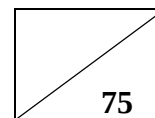
[4 marks]

(b) Hence, find the inflection points of  $f(x)$ .

[3 marks]

**SECTION B** [75 marks]

This section consists of 7 questions. Answer **all** questions.



1. Given  $z = \sqrt{3} + \sqrt{-1}$  such that  $w = \frac{1}{z}$ , where  $w, z \in \mathbb{Z}$ .

Find

- (a) the modulus of  $w$ . [3 marks]

- (b) the argument of  $w$ . [2 marks]

Hence, write  $w$  in polar form. [1 marks]

2. Solve

- (a)  $\ln x + 12 \log_x e = 7$ . [4 marks]

- (b)  $\frac{x^2 + 5}{x - 2} \leq x$ . [5 marks]

3. Given  $h(x) = \frac{2x}{x-2}$ . Defining  $(h \circ h)(x) = h^2(x)$ , determine  $h^2(x)$ . [3 marks]

Hence, deduce  $h^{-1}(x)$  and evaluate  $h^7(8)$ . [4 marks]

4. Functions  $f$  and  $g$ , each with domain  $\mathbb{R}$ , are defined by

$$f : x \rightarrow e^{x+1} \text{ and } g : x \rightarrow |x|.$$

Show that  $(f \circ g)(x) = e^{|x|+1}$ . [2 marks]

- (a) By using algebraic approach, explain why  $(f \circ g)^{-1}(x)$  does not exist.

[3 marks]

- (b) State  $(f \circ g)(x)$  as a piecewise function.

[3 marks]

(c) Given that  $(f \circ g)^{-1}(x)$  exists when the domain for  $(f \circ g)(x)$  is  $\{x: x \geq k\}$ .

(i) State the minimum value of  $k$  and find  $(f \circ g)^{-1}(x)$ . [3 marks]

(ii) Hence, sketch the graph of  $(f \circ g)(x)$  and its inverse on the same axes. [3 marks]

5. (a) Find the horizontal and vertical asymptotes for  $f(x) = \sqrt{\frac{x+1}{x-1}}$ .

[5 marks]

(b) A function  $f(x)$  is defined by

$$f(x) = \begin{cases} \frac{|x|}{x} + a & , \quad x < 2 \\ ax^2 & , \quad 2 \leq x < 3 \\ 1 - bx & , \quad x \geq 3 \end{cases}$$

Determine the values of the constants  $a$  and  $b$  if  $f(x)$  is continuous on the interval  $(0, 3)$  and  $f(5) = 3$ .

[9 marks]

6. (a) Given that  $f(x) = \begin{cases} \frac{4}{3}x^2 & , \quad x \leq 2 \\ 2x + \frac{4}{3} & , \quad x > 2 \end{cases}$  is continuous in the interval  $[0, 4]$ .

By using the definition of  $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ , determine whether  $f'(2)$  exist or not.

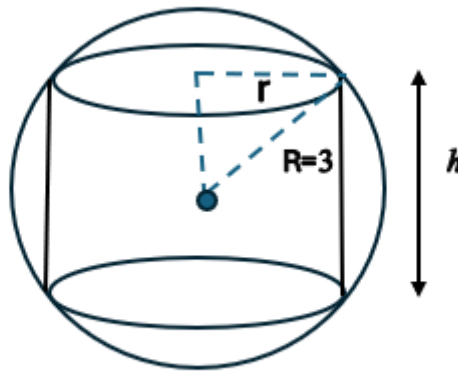
[7 marks]

- (b) Let  $x = \frac{1}{1-t^2}$  and  $y = \frac{1+t^2}{t}$ , where  $t$  is a non-zero parameter.

Show that  $\frac{dy}{dx} = -\frac{1}{2} \left( \frac{1-t^2}{t} \right)^3$ . Hence, find its value at the point  $\left( -\frac{1}{8}, \frac{10}{3} \right)$ .

[9 marks]

7.



The diagram shows a cylinder of radius  $r$  and height  $h$ , inscribed in a sphere of radius  $R = 3$  cm.

- (a) Show that the volume of the cylinder is given by  $V = \pi h \left( 9 - \frac{h^2}{4} \right)$ .

[2 marks]

- (b) Find the height of the cylinder when its volume is at a maximum.

[4 marks]

- (c) Given that the volume of the cylinder is increasing at a rate of  $48 \text{ cm}^3 \text{ s}^{-1}$ .  
Find the rate of change of its height when the height is 2 cm.

[3 marks]

**END OF QUESTION PAPER**